

ULPF.io — Live Application Walkthrough

Universal Log Pre-processing Framework · Standalone Containerized Middleware Gateway

LIVE & VERIFIED

The **ULPF Middleware Gateway** is fully operational. Both the **Spring Boot 3.3.3 (Java 21) backend** and **React 18 + TypeScript control panel** are connected and validated with 100% automated test coverage.

🌐 Live Access Endpoints & Network Listeners

COMPONENT	URL / BINDING	CREDENTIALS / CAPABILITY
Web Control Panel	http://localhost:3000	admin / admin123 (RBAC Protected)
Spring Boot API	http://localhost:8080/api/v1	REST Endpoints (Ingest, Infer, Verify, Export)
Syslog UDP Listener	localhost:1514 (UDP)	Non-blocking Netty Datagram Receiver
Syslog TCP Listener	localhost:1514 (TCP)	Line-framed Netty TCP Receiver
WebSocket Broker	ws://localhost:3000/ws/ulpf	STOMP Telemetry Channels (/topic/metrics, /topic/events)

📸 Visual Verification Across Key Operational Workflows

1. Operations & Normalization Dashboard

REAL-TIME TELEMETRY

Live Events Per Second (EPS) meter, percentile latencies (p50, p95, p99), format distribution (Syslog, CEF, JSON, CSV), and instant sample burst injection.

 **ULPF.io**
MIDDLEWARE GATEWAY

Dashboard

Unknown Log Lab Flagship

Live Stream Viewer

Parser Manager

Rules Configuration

SIEM Exporter

Air-Gapped Gateway

Syslog UDP/TCP: Active (1514)

OCSF Standard: v1.1.0

PIPELINE ACTIVE

UDP/TCP :1514

AIR-GAPPED

2026-08-30 07:27:43 UTC

SOC Security Administrator ROLE_ADMIN

Detected Log Formats

4 Total

CEF	0 (0%)
CSV	0 (0%)
SYSLOG	1 (25%)
CUSTOM	0 (0%)
LEEF	0 (0%)
JSON	1 (25%)
CSV/KV	1 (25%)
UNKNOWN	0 (0%)
CEF/LEEF	1 (25%)

Mapping Confidence Distribution

4 Scored

HIGH (≥90%)

4

Auto-Recommended

MED (70-89%)

0

Review Queue

LOW (<70%)

0

DIQ Escalated

2. Unknown Log Lab — Two-Stage Drain Inference & Operator Approval

FLAGSHIP ENGINE

Mined template of an unseen custom hardware appliance, performed deterministic semantic recognition with **92% High Confidence**, executed operator sign-off, and verified live zero-restart parsing.

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Syslog UDP/TCP: Active (1514)
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PIPELINE ACTIVE

UDP/TCP :1514

AIR-GAPPED

2026-08-30 07:34:23 UTC

SOC Security Administrator
ROLE_ADMIN

<> Live Canonical OCSF JSON Preview

Normalized OCSF v1.1.0 Event (Preview)

Copy

```
{
  "eventId": "PREVIEW-0000-0000-0000",
  "timestampUtc": "2026-08-30T10:32:21Z",
  "event": {
    "category": "network_traffic",
    "type": "custom_inferred",
    "action": "blocked",
    "severity": "high",
    "severityId": 4,
    "message": "Inferred log event from template: SEC-GW-01 <*> <*> <*> <*> <*> <*>"
  },
  "source": {
    "ip": "192.168.10.50:61000"
  },
  "destination": {
    "ip": "10.20.30.40:80"
  }
}
```

✓ Parser Hot-Registered & Active in Memory (Zero-Restart Verification)

The newly approved parser is now live on the gateway. Send another log from this source to verify automated normalization:

SEC-GW-01 2026-08-30T10:32:21Z [BLOCK] proto=TCP src=192.168.10.50 dest=10.20.30.40:80

Test Live Parser

✓ AUTOMATIC LIVE NORMALIZATION SUCCESS! (EVENT ID: ULPF-BE9A90B0-504A-465A-9E4C-30260830103221Z)

3. Cryptographic Lineage & Non-Repudiation Audit

SHA-256 AUDIT

On-demand SHA-256 cryptographic audit recalculates raw hash against database storage on-the-fly and outputs mathematically verified non-repudiation certificates.

Event Cryptographic Audit: ULPF-be9a90b0-504a-465a-9e4c-30c6cb9aeca7

PRE-PARSE SHA-256 DIGEST:
f1a65f4470735854385537dc8ea8f15e5c60ea01c12e6c05891373c94ec48177

Verify Cryptographic Integrity

NON-REPUDIATION VERIFIED

Cryptographic verification SUCCESS. Bit-perfect integrity confirmed across raw and normalized records.

Stored Hash: f1a65f4470735854385537dc8ea8f15e5c60ea01c12e6c05891373c94ec48177

Recomputed: f1a65f4470735854385537dc8ea8f15e5c60ea01c12e6c05891373c94ec48177

Raw Byte Size: 102 bytes • Lineage: custom_2760_parser (v1.0)

Original Bit-Exact Raw Event:

SEC-GW-01 2026-08-30T10:32:21Z [BLOCK] proto=TCP src=192.168.10.50:61000 dst=10.20.30.40:80 user=guest

Canonical OCSF v1.1.0 Event:

Normalized OCSF Representation

Copy

{
 "eventId": "ULPF-be9a90b0-504a-465a-9e4c-30c6cb9aeca7",
 "rawEventId": "RAW-f1a65f4470735854",
 "rawHashSha256": "f1a65f4470735854385537dc8ea8f15e5c60ea01c12e6c05891373c94ec48177",
 "timestampUtc": "2026-08-30T10:32:21Z",
 "event": {
 "category": "network_traffic",
 "type": "firewall",
 "action": "blocked",
 "severity": "high",
 "severityId": 4,
 "details": {
 "src": "192.168.10.50",
 "dst": "10.20.30.40",
 "proto": "TCP",
 "srcPort": 61000,
 "dstPort": 80,
 "user": "guest"
 }
 }
}

4. Multi-SIEM Wire Format Translation

CROSS-SIEM INTEROP

Reverse-translates canonical OCSF telemetry into ArcSight CEF wire, IBM QRadar LEEF wire, and Elastic ECS JSON with automated MITRE ATT&CK technique tagging (T1046).

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PIPELINE
ACTIVE

UDP/TCP
:1514

AIR-
GAPPED

2026-08-30
07:52:32 UTC



SOC Security
Administrator
ROLE_ADMIN

```
"user": {
  "name": "guest"
},
"destination": {
  "ip": "10.20.30.40",
  "port": 80
},
"network": {
  "transport": "TCP",
  "direction": "egress"
},
"observer": {
  "vendor": "DynamicVendor",
  "product": "custom_2760_parser"
},
"threat": {
  "technique": {
    "id": "T1046",
    "name": "Network Service Disc
covery"
  },
  "tactic": {
    "name": "Discovery"
  }
}
}
```

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